MILITARY SPECIFICATION SHEET

GENERATOR SET, DIESEL ENGINE DRIVEN, 30 KILOWATT, 50/60 HERTZ, PRECISE (TACTICAL)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator set described herein shall consist of this document and the issue in effect of Specification MIL-G-52884.

1. SCOPE

1.1 Scope. This specification sheet covers the detail requirement for Type I (Tactical) generator sets, Size 30 kW, Mode I (50/60 Hz), Class 1 (Precise), Model MEP-104A, NSN 6115-00-118-1267.

2. APPLICABLE DOCUMENTS

DRAWINGS

TA 30554-70-104	- Generator Set, Diesel Engine, 30 kw, 50/60 Hz, Tactical Precise, Mode I, Class 1.
30554-79-2001	- Packaging, DOD 30 kW Generator Set.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-G-52884 and as specified herein.

3.2 Description. The set shall be in accordance with top assembly TA 30554-70-104 and as specified herein.

3.2.1 Reliability. The set shall have a reliability of 185 hours (minimum acceptable) Mean-Time-Between-Failure (MTBF).

3.3 Operating speeds. Operating (rated) speed shall be 1,500 rpm for 50 Hz operation and 1,800 rpm for 60 Hz operation.

3.4 Maximum power. The maximum power of the set in kilowatts shall be not less than 37.5 kW.

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3.5 Engine.

3.5.1 Engine horsepower. The minimum continuous brake horsepower of the engine shall be 55 bhp at 1800 rpm.

3.5.2 Engine safety devices. The trip points for engine safety devices shall be as specified in the general specification. The low lubricating oil pressure device shall trip at 17 plus or minus 3 psi.

3.6 Governing system. The electric governing system shall be adjusted such that the set will meet the performance requirements herein.

3.6.1 Frequency regulation. The system shall provide isochronous operation, i.e., for every load change (including rated load) the frequency regulation shall not exceed 1/4 of 1 percent of rated frequency.

3.6.2 Short term stability (30 seconds). At every constant load from no-load to rated load, the system shall maintain frequency within a bandwidth equal to 1/2 of 1 percent of rated frequency. The system shall not permit repetitive periodic frequency variations, commonly called "hunting", even though within the allowable 1/2 of 1 percent band.

3.6.3 Long term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant voltage and any constant load from no-load to rated load, the system shall maintain frequency within a bandwidth equal to 1 percent of rated frequency in a 4 hour operation period.

3.6.4 Transient performance. Following a sudden increase or decrease in load up to and including rated load, the governing system shall reestablish stable engine operating conditions within 1 second. The maximum transient frequency change above or below the new steady-state frequency shall be not more than 1-1/2 percent of rated frequency

3.6.5 Frequency adjustment. By means of the potentiometer on the control cubicle, it shall be possible to adjust the frequency (when the set is operating at any constant load) as follows:

- Between 58 and 62 but not above 65 Hz for the 60 Hz operating condition.
- between 48 and 52 but not below 45 Hz for the 50 Hz operating condition.

It shall not be possible to adjust the frequency for operation within the 48 to 52 Hz range when the frequency selector switch is in the 60 Hz position; and it shall not be possible to adjust for operation in the 58 to 62 Hz range when the frequency selector switch is in the 50 Hz position. Further, it shall not be possible to adjust frequency to a value which will actuate the overspeed protective device.

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3.7 Generator and excitation system.

3.7.1 Transient reactance, rated voltage. The direct axis transient reactance shall be not more than 18 percent.

3.7.2 Negative sequence impedance. The negative sequence impedance shall be not more than 18 percent.

3.7.3 Voltage regulation. The voltage regulation from no-load to rated load and from rated load to no-load shall be not more than 1 perent of rated voltage.

3.7.4 Short term stability (30 seconds). At every constant load from no-load to rated load, the voltage at the set terminals shall remain within a bandwidth equal to 1 percent of rated voltage.

3.7.5 Long term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant frequency and any constant load from no-load to rated load, the voltage shall remain within a bandwidth of 2 percent of rated voltage in a 4 hour operation period.

3.7.6 Transient performance. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows (reactive droop compensation of the voltage regulator being inoperative):

(a) With the set initially operating at no-load, rated voltage, and rated frequency the rms terminal voltage of the set shall not drop to less than 70 percent of no-load voltage (rated voltage) when a balanced 3-phase 0.4 pf (or less, lagging), static load having an impedance of 0.5 per unit is suddenly applied to the output terminals of the set. When connected to the specified load, the output voltage shall recover to a minimum of 95 percent of rated voltage within 0.7 seconds and shall stabilize at or above this voltage. The above specified voltage dip shall not be exceeded when a fully-loaded induction motor of the above specified impedance is used in place of a static load, and no reactions shall be set up to prevent full acceleration of the motor with rated torque applied to its shaft.

(b) When the set is initially operating at rated frequency, rated voltage, and following any sudden change in load from no-load to rated load, the instantaneous rms voltage shall not drop to less than 85 percent of rated voltage and shall reach stable conditions within 0.5 seconds; no overshoot or undershoot of the final voltage may exceed the initial voltage transient in amplitude. The above requirements shall also apply when load is suddenly changed from rated load to no-load, except that the initial voltage transient shall not exceed 115 percent of rated voltage.

(c) The generator set shall be capable of across-the-line starting a motor rated at 1 horsepower per kW of set kW rating.

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3.7.7 Voltage operating range. The set shall operate at any load between no-load and rated load at the following line-to-line voltages:

(a) Between 395 and 480 volts for the 240/416 volt connection and between 197 and 240 volts for the 120/208 volt connection, 60 Hz operation.
(b) Between 380 and 426 volts for the 240/416 volt connection and between 190 and 213 volts for the 120/208 volt connection, 50 Hz operation.

3.7.8 Exciter field current. With the generator temperature stabilized at normal ambient conditions, the exciter field current shall not exceed 1.3 amperes for no-load and 3.5 amperes for rated load at 60 Hz operation and 2.9 amperes for no-load and 5.2 amperes for rated load at 50 Hz operation.

3.8 Parallel operation. With their governor and voltage regulator paralleling circuits properly interconnected and with the voltage across pins A and B of the respective parallel receptacles initially adjusted to an equal value between 4 and 6 volts at rated load, three sets of the same mode and size shall be capable of operating in parallel and dividing load in accordance with the following as system load at rated power factor is varied between zero and 100 percent (and vice versa) of the combined rating of the connected sets:

(a) Real power division. At no time shall the difference between the average individual kilowatt outputs of the sets, when supplying the system load, be greater than 10 percent of the kilowatt rating of one set.
(b) Power exchange. At any constant system load up to the combined rating of the sets in parallel, the power exchange between the sets shall not exceed 4 kW at zero load or 3 kW when operating at loads up to and including 100 percent of the combined rating. Power exchange is the difference between the maximum and minimum power output delivered by one set, for constant system load conditions, as determined from oscillographic measurements.
(c) Reactive power division. At no time shall the difference in average reactive kVA supplied by the sets differ by more than 10 percent of the reactive kVA rating of one set.

(For the above requirements, the initial system load shall be equally divided between the sets, both as to active and reactive components; thereafter, there shall be no adjustments to governors, voltage regulators, or any other component as system load is changed.)

3.9 Protective devices.

3.9.1 Underfrequency. This device shall operate and cause the appropriate malfunction indicator lamp to light when the frequency drops to 55 plus or minus 1 Hz for 60 Hz operation and 46 plus or minus 1 Hz for 50 Hz operation. The device shall not permit closure of the set output circuit interrupter at frequencies below those specified above.

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3.9.2 Permissive parallel relay. This device shall actuate when the signal voltage exceeds 8 plus or minus 1.0 volts rms.

3.9.3 Undervoltage. This device shall operate instantaneously when the voltage drops below 48 volts. It shall not operate instantaneously at voltages above 85 volts. In addition, the device shall operate in 6 plus or minus 2 seconds after the voltage has fallen to and remained at not more than 99 plus or minus 4 volts. Actuation of the undervoltage device shall also cause the appropriate malfunction indicator lamp to light.

3.10 Audio noise. The audio noise sound pressure levels shall not exceed the band pressure levels listed in table "A". Measurements shall be taken with the set operating at any load up to and including rated load. The microphone(s) shall be located 5 feet, 8 inches above ground at positions 25 feet from the perimeter of the set at the center of each side and at each corner of the set (8 positions).

TABLE "A" Commercial frequencies

Octave band Limits (Hz)	Center Frequency (Hz)	Noise Limit (db)
37.5-75	53	89
75-150	106	91
150-300	212	87
300-600	425	83
600-1200	850	79
1200-2400	1700	72
2400-4800	3400	67
4800-9600	6800	63

3.11 Fuel consumption. The maximum fuel consumption shall be such that the set shall operate from the set tank for a minimum period of 8 hours at rated load and frequency without the addition of fuel to the set fuel system.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection of packaging. Inspection of packaging shall be in accordance with drawing 79-2001.

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sted per 4.4.1 Sodium polysulfide test. The stranded samples or annealed copper or copper alloy base material shall be te

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with drawing 79-2001.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Navy - MC
DLA - GS

Preparing activity:
Army - ME

Project 6115-0221

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organic fibrous materials.

1. Purpose. This guideline establishes criteria for the selection and application

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